

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1371

Midrose 138 kV
May 22, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1371. The Interconnection Customer proposes a 72.5 MW battery energy storage facility interconnecting at the Midrose 138 kV substation in the Virginia area. The request was received on January 01, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 448 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,044,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Yulee - Tokoma 138 kV line	Darien - Colesville 138 kV Line	126.8	409	5.4
Arvin - Garden Corners Substation 138 kV	Douglas City - Southwest Research Institute 138 kV	118.9	171	5.9
Arvin - Shuffleton 138 kV	Clark County - Kearneyville 345 kV	117.2	223	6.2
Redbud - Stonewall 138 kV	Springfield - Hp Hood Ct 138 kV	115.6	264	13.0
Freehld - Midrose 138 kV	Lovettsville - West Hills 138 kV	115.4	157	69.7
Douglas City Substation - Garden Corners 138 kV	Douglas City - Southwest Research Institute 138 kV	113.5	214	5.9
Balch - Santa Rosa Energy Ctr 138 kV	Springfield - Hp Hood Ct 138 kV line	111.6	318	13.0
Douglas City - Millille 138 kV L	Clark County - Kearneyville 345 kV	108.6	461	7.4
Sleepy Hollow - Millville 138 kV line	Clark County - Kearneyville 345 kV	108.6	474	7.4
Grimmway-Malaga - Sleepy Hollow 138 kV Line	Clark County - Kearneyville 345 kV L	108.5	487	7.4
W Winchester - International Paper Valliant Ok 138 kV	International Paper Valliant Ok - Kearneyville Substation 345 kV	108.0	427	6.4
Stonewall - 3 Mile Knoll 138 kV L	Springfield - Hp Hood Ct 138 kV	105.9	280	13.0
Balch - Redud Substation 138 kV L	Hp Hood Ct - Springfield 138 kV line	105.0	313	13.0
Arvin - Southwest Research Institute 138 kV	Garden Corners - Douglas City 138 kV	102.8	226	6.5
Gallegos - Killingsworth 138 kV	Tanner - Lcec No. 5 138 kV Line	102.5	354	5.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Redman - Gallegos 138 kV L	Lcec No. 5 - Tanner 138 kV Line	102.4	344	5.8
Redman - Glenbrook Substation 138 kV	Tanner - Lcec No. 5 138 kV	102.3	336	5.8
Glenbrook - Palmyra 138 kV	Tnner - Lcec No. 5 138 kV L	102.2	325	5.8
Palmyra - Grimmway-Malaga 138 kV	Lcec No. 5 - Taner 138 kV	102.0	313	5.8
E Ellensburg - W Hills 138 kV	Kearneyville - Clark County 345 kV line	101.9	282	19.9
Tanner - Gene Substation 138 kV	Gallegos - Killingsworth 138 kV line	101.8	327	6.1
Douglas City - Southwest Research Institute 138 kV	Garden Corners - Douglas City 138 kV Line	101.8	261	6.5
Gene - Grimmway-Malaga 138 kV	Killingsworth Substation - Gallegos Substation 138 kV line	101.6	314	6.1
Terror Lake Microgrid - East Ellensburg 138 kV Line	Clark County - Kearneyville Substation 345 kV	101.6	296	19.9
Tanner - Lcec No. 5 138 kV	Gallegos - Killingsworth 138 kV	101.6	336	6.1
Belleair - Od Chapel 138 kV	Santa Rosa Energy Ctr - Balch Substation 138 kV	100.8	309	10.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tokoma - Yulee 138 kV	Rebuild line with higher-capacity conductor	\$1,071,000	6% (\$59,000)
Arvin - Garden Corners Substation 138 kV	Rebuild line with higher-capacity conductor	\$9,628,000	37% (\$3,588,000)
Arvin - Shuffleton 138 kV	Rebuild line with higher-capacity conductor	\$3,476,000	10% (\$335,000)
Redbud - Stonewall 138 kV	Rebuild line with higher-capacity conductor	\$5,007,000	42% (\$2,084,000)
Midrose - Freehold 138 kV	Rebuild line with higher-capacity conductor	\$9,636,000	59% (\$5,724,000)
Douglas City Substation - Garden Corners 138 kV	Rebuild line with higher-capacity conductor	\$1,324,000	37% (\$494,000)
Balch - Santa Rosa Energy Ctr 138 kV	Rebuild line with higher-capacity conductor	\$7,157,000	42% (\$2,979,000)
Millville - Douglas City 138 kV	Reconductor line	\$7,515,000	10% (\$724,000)
Millville - Sleepy Hollow 138 kV	Reconductor line	\$1,969,000	10% (\$190,000)
Grimmway-Malaga - Sleepy Hollow 138 kV Line	Reconductor line	\$806,000	10% (\$78,000)
W Winchester - International Paper Valliant Ok 138 kV	Reconductor line	\$542,000	45% (\$242,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Stonewall - 3 Mile Knoll 138 kV L	Reconductor line	\$850,000	42% (\$354,000)
Balch - Redud Substation 138 kV L	Reconductor line	\$940,000	42% (\$391,000)
Arvin - Southwest Research Institute 138 kV	Reconductor line	\$3,175,000	37% (\$1,183,000)
Gallegos - Killingsworth 138 kV	Reconductor line	\$1,965,000	51% (\$995,000)
Redman - Gallegos 138 kV	Reconductor line	\$314,000	51% (\$159,000)
Redman - Glenbrook Substation 138 kV	Reconductor line	\$854,000	51% (\$433,000)
Palmyra - Glenbrook 138 kV Line	Reconductor line	\$492,000	51% (\$249,000)
Palmyra - Grimmway-Malaga 138 kV	Reconductor line	\$859,000	51% (\$435,000)
Eat Ellensburg - W Hills 138 kV	Reconductor line	\$966,000	33% (\$322,000)
Tanner - Gene 138 kV	Reconductor line	\$310,000	35% (\$107,000)
Douglas City - Southwest Research Institute 138 kV	Reconductor line	\$867,000	37% (\$323,000)
Gene - Grimmway-Malaga 138 kV	Reconductor line	\$3,621,000	35% (\$1,251,000)
East Ellensburg Substation - Terror Lake Microgrid 138 kV Line	Reconductor line	\$886,000	33% (\$295,000)
Tanner Substation - Lcec No. 5 138 kV	Reconductor line	\$2,722,000	35% (\$941,000)
Old Chapel - Belleair 138 kV	Reconductor line	\$397,000	28% (\$109,000)
	Total allocated to project		\$24,044,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.